

Design a Web Crawler

1. Introduction

A **Web Crawler** is a system used to automatically browse and collect data from the internet.

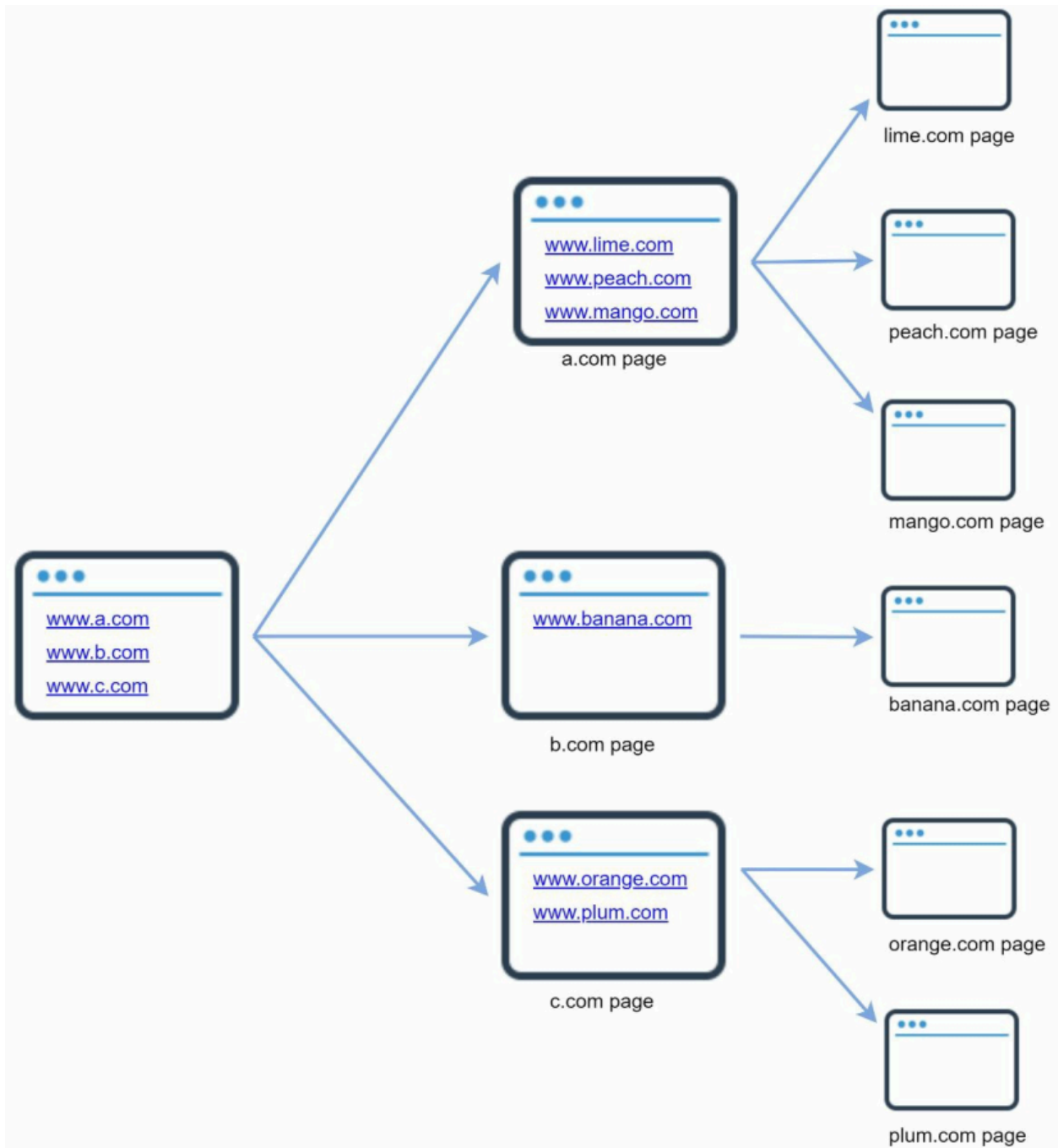
Also known as:

- **Robot**
- **Spider**

2. What Does a Web Crawler Do?

- Starts with a set of initial URLs
- Fetches web pages
- Extracts links from those pages
- Repeats the process recursively

3. Crawling Process (Concept)



Basic flow:

None

Seed URLs → Fetch Page → Extract Links → Add New URLs → Repeat

4. Types of Content Crawled

A web crawler can collect different types of data:

- Web pages (HTML)
- Images
- Videos
- PDF files
- Other digital content

5. Use Cases of Web Crawlers

1. Search Engine Indexing

Most common use case

- Crawlers collect web pages
- Build a **searchable index**

Example:

- Google's crawler (Googlebot)

2. Web Archiving

Purpose:

- Preserve web data for future use

Examples:

- National libraries archive websites

3. Web Mining

Purpose:

- Extract useful insights from web data

Examples:

- Financial firms analyze:
 - Annual reports
 - Shareholder meetings

4. Web Monitoring

Purpose:

- Track illegal or unauthorized content

Examples:

- Detect:
 - Copyright violations
 - Trademark misuse
 - Pirated content

6. System Complexity

Depends on scale:

Small Scale

- Simple implementation
- Few URLs
- Can be built quickly (hours/days)

Large Scale

- Handles billions of pages

- Requires:
 - Distributed systems
 - Fault tolerance
 - Continuous updates

7. Key Takeaway

- Web crawler is a **foundation system** for:
 - Search engines
 - Data mining
 - Monitoring systems
- Complexity increases significantly with scale